

RAB PHI & ITSM-HI

Enterprise-Grade Developer Handover Pack

Final authoritative technical design, architecture, logic, questions, validations and error handling

1. Executive Lock

This document is the single source of truth for the PHI and ITSM-HI platform build. It contains the locked terminology, site architecture, framework structures, all questions, exact weights, scoring formulas, scope-creep treatment, confidence logic, controlled fields, validations, error handling, API contract, data model, dashboard/report design, AI payloads and implementation pseudocode. No silent reduction, renaming, simplification or question downgrade is allowed without formal change control.

2. Platform Scope and Two-Level Model

- PHI (Programme Health Index) for ERP, digital transformation and most transformation or delivery programmes.
- ITSM-HI (IT Service Management Health Index) for service operations, transition, control maturity and BAU readiness.
- Snapshot = public quick assessment, lead capture, instant board-level result.
- Full = consultant-led deep assessment, evidence capture, board-grade report.

Framework	Term	Snapshot Questions	Full Questions
PHI	Pillars	18	75
ITSM-HI	Domains	22	65

3. Site Map and Product Architecture

Public layer:

- Home
- PHI Snapshot Assessment
- ITSM-HI Snapshot Assessment
- Snapshot Result Page
- Book Consultation CTA

Authenticated consultant layer:

- Consultant Dashboard
- Assessment List
- PHI Full Assessment
- ITSM-HI Full Assessment
- Report Builder / Review
- Lead / CRM View
- Configuration (admin only)
- Audit / Prompt Logs (admin only)

4. Exact Technical Architecture

Layer	Preferred Implementation	Purpose
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Frontend	React / Next.js	Public forms, consultant UI, dashboards, reports
Backend	Laravel preferred, or Node.js / TypeScript	Business logic, scoring engine, workflows, auth, APIs
Database	PostgreSQL	Question bank, configs, assessments, responses, results, prompts, reports, audit
AI Layer	OpenAI 4.1	Structured narrative generation
Report Rendering	HTML templates	Board-grade dashboard/report layout
PDF Generation	Puppeteer or Playwright	Reliable HTML to PDF export
Storage	Object storage or local environment storage	PDFs and exported files

5. Locked Framework Structures

5.1 PHI final pillars and weights

Code	Pillar	Weight	Critical
P1	Governance & Decision-Making	1.5	Yes
P2	Planning, Scope & Delivery Control	1.5	Yes
P3	Business Alignment, Value & Financial Control	1.4	Yes
P4	Change, Training & Adoption	1.2	No
P5	Data Readiness & Migration	1.2	No
P6	Solution, Architecture & Process Fit	1.2	No
P7	Cutover & Go-Live Readiness	1.3	Yes
P8	Delivery Capability, Vendor & Resourcing	1.2	No
P9	Operational & Automation Readiness	1.0	No

5.2 ITSM-HI final domains and weights

Code	Domain	Weight	Critical
D1	Service Governance & Ownership	1.4	Yes
D2	Incident & Major Incident Management	1.4	Yes
D3	Service Request Management	1.0	No
D4	Problem Management	1.2	No
D5	Change & Release Management	1.3	Yes
D6	Service Performance, SLA & Reporting	1.2	No
D7	Service Transition & BAU Readiness	1.2	Yes
D8	Service Operations & Support Model	1.2	No

D9	Supplier & Vendor Service Management	1.0	No
D10	Operational Resilience & Continuity	1.3	Yes
D11	Automation, Tooling & Service Optimisation	1.0	No

6. Exact Question Sets

6.1 PHI Snapshot Questions

ID	Pillar	Question
P1_S1	P1	Decision-making and escalation are clear and effective.
P1_S2	P1	Programme reporting reflects the true position.
P2_S1	P2	The integrated plan is realistic, credible and actively controlled.
P2_S2	P2	Scope is baselined and change is governed without destabilising delivery.
P3_S1	P3	Business objectives, benefits and ownership are clearly defined.
P3_S2	P3	Budget, cost exposure and value delivery are visible and controlled.
P4_S1	P4	Users and business leaders are prepared and engaged.
P4_S2	P4	Training and adoption readiness are realistic and measurable.
P5_S1	P5	Data quality and migration readiness are understood.
P5_S2	P5	Data ownership and validation responsibilities are clear.
P6_S1	P6	The solution and architecture fit the business and technical needs.
P6_S2	P6	Process design and customisation are controlled and proportionate.
P7_S1	P7	Cutover and go-live readiness are on track.
P7_S2	P7	Critical dependencies and contingency planning are credible.
P8_S1	P8	The programme has the right capability, vendor control and capacity to deliver.
P8_S2	P8	Key delivery roles and accountabilities are covered.
P9_S1	P9	BAU readiness, support and automation opportunities are understood.
P9_S2	P9	Operational reporting and support handover are feasible.

6.2 PHI Full Questions

ID	Pillar	Question
P1_F1	P1	Sponsor authority is clear and active.
P1_F2	P1	Steering decisions are timely and evidence-based.
P1_F3	P1	Escalation routes are defined and used.
P1_F4	P1	RAID governance is disciplined and current.
P1_F5	P1	Programme roles and responsibilities are explicit.
P1_F6	P1	Reporting reflects the true delivery position.
P1_F7	P1	Governance forums drive action rather than status-only discussion.
P1_F8	P1	Decision logs and action ownership are maintained.
P2_F1	P2	The integrated plan is baselined and actively managed.
P2_F2	P2	Critical path and dependencies are visible and controlled.
P2_F3	P2	Milestones are realistic and measurable.
P2_F4	P2	Forecasting and replanning are disciplined.
P2_F5	P2	Risks and issues are actively managed.
P2_F6	P2	Scope baseline is defined and understood.
P2_F7	P2	Change requests are formally assessed and approved.
P2_F8	P2	Change impact on time, cost and value is assessed before approval.
P2_F9	P2	Change volume is not destabilising delivery.
P3_F1	P3	Business case assumptions remain valid.
P3_F2	P3	Benefits are clearly defined and owned.
P3_F3	P3	KPIs for success are measurable.
P3_F4	P3	Budget tracking is accurate and current.
P3_F5	P3	Forecast cost at completion is understood.
P3_F6	P3	Change costs are controlled and visible.
P3_F7	P3	Value delivery is reviewed with the business.
P3_F8	P3	Business ownership is active and accountable.
P4_F1	P4	Stakeholder impacts are understood.
P4_F2	P4	Change strategy is clear and realistic.
P4_F3	P4	Training plans are role-based and practical.
P4_F4	P4	Training attendance and readiness are measured.

P4_F5	P4	Business leaders are engaged and visible.
P4_F6	P4	Resistance points are known and managed.
P4_F7	P4	Change champions are in place where needed.
P4_F8	P4	Adoption measures exist beyond training completion.
P5_F1	P5	Data migration strategy is clear.
P5_F2	P5	Data ownership is assigned.
P5_F3	P5	Data quality issues are visible and managed.
P5_F4	P5	Data cleansing is planned and resourced.
P5_F5	P5	Mock migrations are planned or performed.
P5_F6	P5	Reconciliation controls are defined.
P5_F7	P5	Business validation of migrated data is credible.
P5_F8	P5	Migration cutover dependencies are understood.
P6_F1	P6	Solution design supports the target operating model.
P6_F2	P6	Process design is documented and agreed.
P6_F3	P6	Customisation is controlled and justified.
P6_F4	P6	Integration complexity is understood.
P6_F5	P6	Architecture choices are scalable and supportable.
P6_F6	P6	Design gaps are identified and owned.
P6_F7	P6	Workarounds are acceptable and controlled.
P6_F8	P6	Non-functional requirements are understood where relevant.
P7_F1	P7	Cutover plan is detailed and credible.
P7_F2	P7	Cutover roles and ownership are clear.
P7_F3	P7	Cutover rehearsal strategy exists.
P7_F4	P7	Go / no-go criteria are defined.
P7_F5	P7	Rollback or fallback planning is credible.
P7_F6	P7	Day-1 operational processes are ready.
P7_F7	P7	Hypercare model is agreed.
P7_F8	P7	Support escalation model is ready.
P7_F9	P7	Critical business dependencies are accounted for.
P7_F10	P7	Business sign-off readiness is realistic.
P8_F1	P8	Programme leadership capability is sufficient.
P8_F2	P8	PMO support is effective.
P8_F3	P8	Internal resource capacity is realistic.
P8_F4	P8	Key skill gaps are known and mitigated.
P8_F5	P8	Vendor or SI performance is governed effectively.
P8_F6	P8	Commercial accountabilities are clear.

P8_F7	P8	Cross-team collaboration supports delivery.
P8_F8	P8	Team stability is sufficient for the current phase.
P9_F1	P9	BAU support model is defined.
P9_F2	P9	Operational handover requirements are known.
P9_F3	P9	Support documentation and knowledge transfer are planned.
P9_F4	P9	Automation opportunities are identified and prioritised.
P9_F5	P9	Operational reporting requirements are clear.
P9_F6	P9	Service acceptance into BAU is credible.
P9_F7	P9	Control ownership after go-live is clear.
P9_F8	P9	Ongoing optimisation approach is defined.

6.3 ITSM-HI Snapshot Questions

ID	Domain	Question
D1_S1	D1	Service ownership and accountability are clear.
D1_S2	D1	Governance and review cadence are effective.
D2_S1	D2	Incidents are managed in a controlled and timely manner.
D2_S2	D2	Major incident decision-making is effective.
D3_S1	D3	Requests follow a clear and efficient fulfilment path.
D3_S2	D3	Request categorisation and routing are controlled.
D4_S1	D4	Repeat issues are identified and addressed.
D4_S2	D4	Root cause resolution is pursued beyond symptom treatment.
D5_S1	D5	Change control is effective and risk-managed.
D5_S2	D5	Release decisions are disciplined and supported by evidence.
D6_S1	D6	Service performance and SLAs are visible.
D6_S2	D6	Reporting drives action rather than passive review.
D7_S1	D7	Transition into BAU is prepared and controlled.
D7_S2	D7	Operational acceptance criteria are defined.
D8_S1	D8	The support model is clear and workable.
D8_S2	D8	Escalation and handoffs are defined.
D9_S1	D9	Suppliers are governed against service obligations.

D9_S2	D9	Vendor performance is visible and challenged.
D10_S1	D10	Continuity and resilience controls are in place.
D10_S2	D10	Operational resilience risks are known and managed.
D11_S1	D11	Tooling supports efficient service delivery.
D11_S2	D11	Automation is used or planned where it adds control and efficiency.

6.4 ITSM-HI Full Questions

ID	Domain	Question
D1_F1	D1	Service ownership is formally assigned.
D1_F2	D1	Service governance forums are active and effective.
D1_F3	D1	Roles and responsibilities are understood.
D1_F4	D1	Governance decisions lead to measurable action.
D1_F5	D1	Service risks are visible to the right stakeholders.
D1_F6	D1	Ownership of service controls is clear.
D2_F1	D2	Incident logging and categorisation are controlled.
D2_F2	D2	Incident prioritisation is consistent.
D2_F3	D2	Major incident process is defined and used.
D2_F4	D2	Escalation during incidents is timely.
D2_F5	D2	Restoration targets are realistic and monitored.
D2_F6	D2	Incident trends are reviewed and acted upon.
D3_F1	D3	Request models are defined and maintained.
D3_F2	D3	Request fulfilment responsibilities are clear.
D3_F3	D3	Request backlog and ageing are visible.
D3_F4	D3	Standard requests are automated where appropriate.
D3_F5	D3	Request quality and user experience are reviewed.
D4_F1	D4	Problem backlog is visible and prioritised.
D4_F2	D4	Known error handling is established.
D4_F3	D4	Problem investigations address root cause.
D4_F4	D4	Problem actions are tracked to closure.
D4_F5	D4	Repeat incident patterns drive problem records.
D5_F1	D5	Change assessment is disciplined.
D5_F2	D5	Change approval authority is clear.
D5_F3	D5	Emergency change handling is controlled.

D5_F4	D5	Release and deployment planning is effective.
D5_F5	D5	Change success/failure is reviewed.
D5_F6	D5	Change risk is evaluated before implementation.
D5_F7	D5	Forward schedule of change supports coordination.
D6_F1	D6	SLA definitions are clear and realistic.
D6_F2	D6	Performance data is accurate.
D6_F3	D6	Reporting is timely and useful.
D6_F4	D6	Service reviews drive corrective action.
D6_F5	D6	Service KPIs reflect actual user/service outcomes.
D6_F6	D6	Capacity or performance risks are visible.
D7_F1	D7	Service transition planning is defined.
D7_F2	D7	Operational acceptance criteria are applied.
D7_F3	D7	Knowledge transfer into BAU is complete.
D7_F4	D7	Support readiness is validated before handover.
D7_F5	D7	Transition risks are managed.
D7_F6	D7	Hypercare or early life support is planned.
D8_F1	D8	Support model roles and queues are clear.
D8_F2	D8	Handoffs between support levels are controlled.
D8_F3	D8	Operational work instructions are available.
D8_F4	D8	Escalation paths are understood.
D8_F5	D8	Shift / coverage model is fit for purpose.
D8_F6	D8	Operational issues are reviewed and improved.
D9_F1	D9	Supplier obligations are documented.
D9_F2	D9	Supplier performance is monitored.
D9_F3	D9	Vendor governance meetings are effective.
D9_F4	D9	Escalation with suppliers is controlled.
D9_F5	D9	Third-party dependencies are understood.
D10_F1	D10	Continuity plans are defined.
D10_F2	D10	Recovery responsibilities are clear.
D10_F3	D10	Resilience risks are assessed.
D10_F4	D10	Resilience controls are tested where appropriate.
D10_F5	D10	Single points of failure are understood.
D10_F6	D10	Continuity arrangements align with service criticality.
D11_F1	D11	Core tooling supports service management processes.
D11_F2	D11	Tool configuration supports accurate control and reporting.
D11_F3	D11	Automation is used to reduce manual

		risk and effort.
D11_F4	D11	Improvement opportunities are captured and prioritised.
D11_F5	D11	Tooling data quality is sufficient for reporting and control.

7. Scoring Scale, Weighting and Status Logic

Score	Meaning
1	Critical / not in place / failing
2	Weak / inconsistent
3	At Risk / partially in place
4	Controlled / mostly effective
5	Strong / fully effective

Question weighting is equal within each pillar or domain. Weighting is applied only at pillar/domain level.

`raw_pillar_or_domain_score = ROUND( SUM(answered question scores in pillar/domain) / COUNT(answered question scores in pillar/domain), 2 )`

`weighted_pillar_or_domain_score = ROUND( raw_score * weight, 2 )`

`overall_score = ROUND( SUM(all weighted pillar/domain scores) / SUM(all configured weights for scored pillars/domains), 2 )`

Condition	Overall Status
<code>overall_score < 3.00</code>	Critical
<code>overall_score >= 3.00 and overall_score < 4.00</code>	At Risk
<code>overall_score >= 4.00</code>	Controlled

8. Explicit Scope Creep / Scope Stability Design (PHI)

Scope creep must be explicit and visible. It must not remain a hidden interpretation across multiple answers.

Question ID	Coverage
P2_S2	Snapshot scope baseline and change governance signal
P2_F6	Scope baseline is defined and understood
P2_F7	Change requests are formally assessed and approved
P2_F8	Change impact on time, cost and value is assessed before approval
P2_F9	Change volume is not destabilising delivery
P3_F4	Budget tracking is accurate and current
P3_F6	Change costs are controlled and visible

`scope_stability_index = ROUND( AVG( P2_F6, P2_F7, P2_F8, P2_F9, P3_F4, P3_F6 ), 2 )`

Condition	Scope Status
<code>scope_stability_index < 3.00</code>	Unstable
<code>scope_stability_index >= 3.00 and scope_stability_index < 4.00</code>	Under Pressure
<code>scope_stability_index >= 4.00</code>	Controlled

9. Confidence Model

9.1 Snapshot confidence

Snapshot confidence is completion-based only.

$\text{completion_percent} = \text{ROUND}(\text{answered_questions} / \text{total_configured_questions}, 2)$

Condition	Snapshot Confidence
$\text{completion_percent} < 0.60$	Low
$\text{completion_percent} \geq 0.60$ and $\text{completion_percent} < 0.85$	Medium
$\text{completion_percent} \geq 0.85$	High

9.2 Full confidence

Full confidence must combine completion, evidence source quality and assessor confidence.

$\text{evidence_score} = \text{AVG}(\text{evidence source mapped values})$

$\text{assessor_confidence_score} = \text{AVG}(\text{assessor confidence mapped values})$

$\text{full_confidence_index} = \text{ROUND}((\text{completion_percent} * 0.30) + (\text{evidence_score} * 0.35) + (\text{assessor_confidence_score} * 0.35), 2)$

Mapped Values	Value
Evidence Source: Interview	0.60
Evidence Source: Observation	0.70
Evidence Source: Document	0.85
Evidence Source: System Evidence	1.00
Assessor Confidence: Low	0.40
Assessor Confidence: Medium	0.70
Assessor Confidence: High	1.00

Condition	Full Confidence
$\text{full_confidence_index} < 0.60$	Low
$\text{full_confidence_index} \geq 0.60$ and $\text{full_confidence_index} < 0.85$	Medium
$\text{full_confidence_index} \geq 0.85$	High

10. CRM-Ready Controlled Input Fields

Field	Control	Rule
Client	Free text	Mandatory
Programme / Service	Free text	Mandatory
Type	Dropdown	Mandatory
Subtype	Dropdown (future optional)	Optional
Assessor	Free text	Mandatory on full, optional on snapshot
Role	Dropdown	Mandatory
Date	Date field	Mandatory
Score	Dropdown 1 to 5 only	Mandatory where question answered
Evidence Source	Dropdown	Mandatory on full assessments
Assessor Confidence	Dropdown	Mandatory on full assessments

Type dropdown values:

- ERP
- Digital Transformation
- ITSM / Service
- Data / AI
- Cyber Security
- M&A / Integration
- Transformation / Programme
- Other

Role dropdown values:

- Sponsor
- Director
- Programme Manager
- Project Manager
- IT Lead
- Service Owner
- Business Lead
- Other

11. Dashboard and Report Outputs

- Overall Score
- Overall Status
- Confidence
- Priority / Flag
- PHI only: Scope Status
- Heatmap by pillar/domain
- Radar / spider chart
- Ranked bar chart from weakest to strongest
- Top 3 Risks
- Top 3 Strengths
- Executive Summary
- Full assessments also include RCA preview and priority actions preview

Report Section	Snapshot	Full
Client & Assessment Context	Yes	Yes
Executive Summary	Yes	Yes
Overall Score / Status / Confidence	Yes	Yes
Scope Status (PHI)	Yes	Yes
Top Risks	Yes	Yes
Top Strengths	Yes	Yes
Board Commentary	Yes	Yes
Immediate Actions	Yes	Yes
Root Cause Analysis	No	Yes
Likely Implications	No	Yes
Priority Actions	No	Yes
30/60/90 Day Plan	No	Yes
OpenAI 4.1 Prompt Payload	Yes	Yes

12. Exact Data Model

Entity	Key Fields
framework	id, code, name, is_active
pillar_domain	id, framework_id, code, name, weight, is_critical, display_order
question	id, framework_id, pillar_domain_id, assessment_level, question_code, question_text, is_active, display_order
assessment	id, framework_id, level, client_name, programme_service_name, type, subtype, assessor_name, role, created_at, updated_at, status
assessment_response	id, assessment_id, question_id, numeric_score, notes_text, evidence_source, assessor_confidence, assessor_comment
pillar_domain_result	assessment_id, pillar_domain_id, raw_score, weight, weighted_score, rank_low, rank_high, heatmap_band
assessment_result	assessment_id, overall_score, overall_status, confidence, priority_flag, scope_status, scope_stability_index
report_output	assessment_id, report_type, html_output, json_output, pdf_path
prompt_log	assessment_id, prompt_type, prompt_payload_json, generated_response_json, model_name, created_at
audit_log	id, entity_type, entity_id, action, before_json, after_json, actor, created_at
config	config_key, config_value_json, version

13. Exact API Contract

Method	Endpoint	Purpose
GET	/api/frameworks	Return framework metadata
GET	/api/frameworks/{framework}/questions?snapshot=true	Return snapshot question set
GET	/api/frameworks/{framework}/questions?full=true	Return full question set
POST	/api/assessments/snapshot	Create snapshot and save responses
POST	/api/assessments/full	Create/update full assessment and save responses
POST	/api/assessments/{id}/calculate	Run scoring engine
POST	/api/assessments/{id}/generate-report	Build report payload and call AI
POST	/api/assessments/{id}/generate-pdf	Render HTML report to PDF
GET	/api/assessments/{id}/results	Return dashboard and result payload
GET	/api/assessments/{id}/report	Return report sections and prompt outputs
GET	/api/config	Return current scoring/config values

14. Validation Rules

Area	Rule	Behaviour
Score	Must be dropdown 1 to 5 only	Reject invalid values
Type	Must be from dropdown list	Reject invalid values
Role	Must be from dropdown list	Reject invalid values
Snapshot completion	Minimum completion required to generate output	If below threshold show incomplete / low confidence

Full assessment	Allow draft save but block finalisation if required fields missing	Show validation summary
Evidence source	Mandatory on full assessments	Reject blank on finalisation
Assessor confidence	Mandatory on full assessments	Reject blank on finalisation
Question mapping	Every question must map to a valid pillar/domain	Fail configuration and log error

15. Error Handling, Logging and Recovery

The build must include explicit error handling and auditability. This is non-negotiable.

Error Type	Detection	Handling
Invalid score value	Validation layer	Reject request with field-level error
Invalid type or role	Validation layer	Reject request with field-level error
Missing evidence or assessor confidence on full assessment	Validation layer	Block finalisation; allow draft save
Question not mapped to valid pillar/domain	Config/scoring layer	Fail scoring, log config error, notify admin
Division by zero / no scored answers	Scoring layer	Return null/incomplete status; do not crash
AI generation failure	AI service layer	Persist results, store failed prompt, allow retry
PDF generation failure	Report service	Persist HTML output, log failure, allow regeneration
Config version mismatch	Deployment/QA layer	Block release or mark environment invalid

- All calculation runs must be stored against assessment id and scoring version.
- All prompt payloads must be stored before AI submission.
- All final reports must be reproducible from saved responses and config version.
- No error should silently degrade the score or report output.

16. Structured AI Payload Design

Snapshot payload:

```
{
  "framework": "PHI or ITSM-HI",
  "level": "snapshot",
  "overall_score": 3.64,
  "status": "At Risk",
  "confidence": "High",
  "priority_flag": "High",
  "scope_status": "Under Pressure",
  "top_risks": ["..."],
  "top_strengths": ["..."]
}
```

Full payload:

```
{
  "framework": "PHI or ITSM-HI",
  "level": "full",
  "overall_score": 3.42,
  "status": "At Risk",
  "confidence": "Medium",
  "priority_flag": "Medium",
  "scope_status": "Under Pressure",
  "top_risks": ["..."],
  "top_strengths": ["..."]
}
```

```

    "priority_flag": "High",
    "scope_status": "Under Pressure",
    "scope_stability_index": 3.10,
    "pillar_domain_scores": [{ "code": "P1", "name": "Governance & Decision-Making", "raw_score": 2.8 }],
    "top_risks": [ "... "],
    "top_strengths": [ "... "],
    "assessor_notes": "...",
    "evidence_summary": "...
  }

```

Required generated sections:

- Executive Summary
- Overall Health / Status
- Scope & Change Analysis for PHI
- Top Risks
- Top Strengths
- Board Commentary
- Root Cause Analysis for full assessments
- Likely Implications for full assessments
- Priority Actions
- 30/60/90 Day Plan

17. Developer Calculation Pseudocode

```

validate metadata and responses
group responses by pillar/domain
for each pillar/domain:
    raw_score = round(sum(scores)/count(scores), 2)
    weighted_score = round(raw_score * weight, 2)
    heatmap_band = Red if raw_score < 2.5 else Amber if raw_score < 3.8 else Green
overall_score = round(sum(weighted_scores)/sum(weights for scored pillars/domains), 2)
status = Critical if overall_score < 3 else At Risk if overall_score < 4 else Controlled
if level == snapshot:
    confidence = completion-based
else:
    confidence = completion + evidence + assessor confidence weighted model
if framework == PHI:
    scope_stability_index = average([P2_F6, P2_F7, P2_F8, P2_F9, P3_F4, P3_F6])
    scope_status = Unstable / Under Pressure / Controlled
top_risks = bottom 3 by raw score
top_strengths = top 3 by raw score
persist results

```

18. Build Tickets / Work Packages

Work Package	Key Deliverables
WP1 Foundation	Project setup, auth, config tables, base API, audit logging
WP2 Question Engine	Frameworks, pillars/domains, question bank, controlled metadata fields
WP3 Snapshot Flow	Public forms, validation, scoring, lead capture, summary output
WP4 Full Assessment Flow	Consultant workflow, evidence capture, confidence inputs, draft/final states
WP5 Scoring Engine	Weighted scoring, status logic, scope stability,

	risks/strengths
WP6 Dashboards	Heatmap, radar, ranked bar, executive KPI blocks
WP7 Reports	HTML board reports, AI narrative integration, RCA/actions/30-60-90
WP8 PDF	HTML to PDF via Puppeteer or Playwright
WP9 QA & Regression	Validation suite, scoring parity tests, config/version tests, report generation tests

19. Acceptance Criteria

No.	Criterion
1	All questions listed in this TDD exist in the application exactly once with the same IDs and wording.
2	PHI uses 9 pillars and ITSM-HI uses 11 domains.
3	PHI snapshot has 18 questions, PHI full has 75, ITSM snapshot has 22, ITSM full has 65.
4	PHI explicitly surfaces scope stability and scope status.
5	Scores accept only 1 to 5.
6	Dashboard includes heatmap, radar and ranked bar chart.
7	Snapshot and full board reports render from the same calculation engine.
8	Controlled fields are dropdown-based for CRM cleanliness.
9	PDF is rendered from HTML using the same report data.
10	No downgrade or silent simplification is permitted without formal change control.

20. Final Lock Statement

This document is the enterprise-grade developer handover pack and the single source of truth for the build. It includes all pillars, all domains, all question sets, explicit scope-creep coverage, final weights, formulas, confidence logic, architecture, validations, error handling, data model, API, dashboard design, report design and AI payload design. It is intended to be handed directly to the development team.